

In this project, you will construct artificial datasets to investigate how different properties of a dataset affect the performance of certain linear regression and classification methods.

Please hand in a Jupyter notebook with your Python or Julia code, along with brief explanations and discussions for each subquestion. The machine learning methods should be implemented from scratch. You may, however, use standard libraries such as NumPy and SciPy for linear algebra and other basic numerical operations. Please ensure that all code in the notebook runs without modification.

The idea of this project is that you experiment with the data and algorithms yourselves. You can use generative AI, but if you do, please make sure that you stay in charge and that you check (and discuss among yourselves) the LLM-generated material. All group members should fully agree with and understand everything you hand in. In the presentation and interview you should be able to answer detailed questions and be able to motivate the choices you made.

Use Brightspace to hand in your work. The submission hand-in date for this project is 2 November 2026. Please also prepare a short presentation (10 minutes) in which you discuss the experiments you performed and the results you found particularly interesting (there is not enough time to discuss everything). The presentations and interviews will take place on 4 and 5 November 2026 and will last no more than 30 minutes in total.

1. (a) Generate a 300×20 data matrix X , where each entry is uniformly random. Generate an outcome vector y , which is a linear combination of the columns of X with uniformly random weights, and some Gaussian noise added to each entry of y .
(b) Write a function to divide the data set into a train and test sets.
(c) Write functions for OLS and ridge regression and apply them to your synthetic data set. Discuss the performance on train and test sets.
(d) Create a data matrix with many multicollinearities by adding a large number (say, 200) columns to X that are linear combinations of the original 20 columns with some Gaussian noise added to each entry. Run OLS and ridge regression and discuss the performance on train and test sets. Is it hard to find a good value for λ ?
(e) Now instead of adding multicollinearities, add many irrelevant feature columns to X which have no relation to the outcome vector y . Again run OLS and ridge regression and discuss the performance on train and test sets.
2. (a) Implement a function for logistic regression (with regularization) as in the lecture.
(b) Create a random data matrix X and construct an output vector y by generating a random weight vector w and setting $y_i = \text{sign}(x_i^\top w)$, where $x_i^\top$ is the i -th row of X . Use a test/train split and check the performance of ridge and logistic regression for binary classification. Do you see a large difference in performance between these methods?
(c) Now create a data set (X, y) for binary classification (with $X \in \mathbb{R}^{n \times d}$ and $y \in \{-1, 1\}^n$) such that, given a test/train split, OLS and ridge perform very badly but logistic regression performs well. What kind of properties of your data set are responsible for this?
3. Double descent is a phenomenon where the test error first decreases as model complexity increases, then increases sharply near the interpolation threshold (where the model is just able

to perfectly fit the training data), and then decreases again as model complexity increases further. For neural networks, the phenomenon is not yet fully understood.

For OLS, we can investigate double descent by varying the number of data points while keeping the number of features fixed. (This is called sample-wise double descent.) Construct synthetic data to investigate this phenomenon, and use the minimum-norm OLS solution when there are fewer data points than features. Plot the training and test errors as functions of the number of training data points. Which properties of your dataset influence the phenomenon? Explain the behaviour you observe.